

Improving Urban Vegetation Segmentation: A Cross-City Evaluation Using Aerial Imagery

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Abstract—The segmentation of urban vegetation from aerial imagery is important for environmental monitoring and urban planning. Although deep learning models achieve strong performance on benchmark datasets, their ability to generalize across cities remains limited due to domain shift. This study evaluates cross-city generalization using a U-Net model with a ResNet-34 encoder trained on the ISPRS Potsdam dataset and tested on unseen QGIS-derived datasets from Bamberg, Erlangen, and Potsdam without domain adaptation. Performance is evaluated using Intersection over Union (IoU), pixel accuracy, qualitative analysis, and confusion matrices. In addition, a shadow-based analysis using HSV thresholding is conducted to assess the impact of illumination variations on segmentation performance. The results show substantial performance degradation across cities, particularly in dense urban environments with complex land-cover patterns. Vegetation segmentation performance decreases as urban density increases, while shadows introduce added localized segmentation difficulty. These findings indicate that structural complexity and domain shift are significant challenges for cross-city generalization.

Index Terms—semantic segmentation, urban vegetation mapping, cross-city generalization, domain shift, aerial imagery, deep learning, remote sensing

I. INTRODUCTION

Urban vegetation mapping from aerial imagery is important for environmental monitoring, urban planning, and climate analysis. Recent advances in deep learning have improved semantic segmentation performance in high-resolution remote sensing imagery. However, despite strong results on benchmark datasets, segmentation models often struggle to generalize across cities with different urban layouts, vegetation distributions, and imaging conditions. This limits the reliability of deploying segmentation models in real-world urban environments.

Encoder-decoder architectures such as U-Net, DeepLabV3+, and PSPNet have demonstrated strong performance in the semantic segmentation of aerial imagery by learning hierarchical spatial representations from high-resolution data [1]–[3]. More recently, transformer-based approaches have improved segmentation accuracy through enhanced global contextual modelling [4]. Benchmark datasets such as ISPRS Potsdam have enabled systematic evaluation of these models for urban land-cover classification tasks[5]. Despite these advances, most studies primarily evaluate performance within the same

dataset distribution, where training and testing data share similar spatial and radiometric characteristics.

In practical deployment scenarios, segmentation models are often applied to unseen cities that differ largely from the training domain. Variations in urban density, land-cover composition, annotation structure, and illumination conditions introduce domain shift, leading to significant performance degradation. While prior work has explored domain adaptation and transfer learning approaches to address this issue, many methods require access to target-domain data or supplementary retraining. Additionally, illumination variations, such as shadows, can increase ambiguity in dense urban environments by altering spectral characteristics and obscuring object boundaries [6]. This raises an important question: What factors most strongly influence segmentation failure under cross-city domain transfer?

This study investigates source-only cross-city generalization for urban vegetation segmentation using aerial imagery. A U-Net model with a ResNet-34 encoder is trained on the ISPRS Potsdam dataset and evaluated on unseen QGIS-derived datasets from Bamberg, Erlangen and Potsdam without domain adaptation or fine-tuning. The primary objective is to analyse how urban density, structural complexity, and radiometric variation affect segmentation robustness across cities. In addition, a shadow-based analysis is conducted to evaluate the localized impact of illumination variation on segmentation performance. The study provides three main contributions: 1) a source-only evaluation of cross-city segmentation generalization across multiple urban domains, 2) analysis of urban density and class imbalance as drivers of performance degradation, and 3) investigation of shadow effects under cross-domain conditions.

II. RELATED WORK

Deep learning has considerably advanced semantic segmentation in remote sensing imagery, especially for urban land-cover analysis. Early approaches to convolutional neural networks demonstrated that fully convolutional architectures could effectively learn spatial features from high-resolution aerial imagery [7]. Encoder-decoder architectures such as U-Net improved dense prediction performance through multi-scale feature aggregation and precise spatial localization [1]. Subsequent methods, including DeepLab-based architectures

and PSPNet, introduced atrous convolution and pyramid pooling strategies to improve contextual understanding and segmentation accuracy in complex scenes [2], [3]. More recently, transformer-based segmentation models have demonstrated improved global contextual modelling and long-range feature representation in dense prediction tasks [4].

Benchmark datasets such as ISPRS Potsdam have enabled systematic evaluation of semantic segmentation models for urban aerial imagery [5]. Previous studies have reported strong in-domain performance for land-cover classes including buildings, vegetation, roads, and impervious surfaces [8], [9]. However, many existing evaluations are performed within the same dataset distribution, where training and testing samples share similar spatial structure, annotation style, and radiometric characteristics. As a result, the reported benchmark performance often does not reflect real-world deployment conditions.

Cross-domain and cross-city generalization remain major challenges in aerial image segmentation. Models trained on one geographic region frequently experience substantial performance degradation when applied to unseen cities due to differences in urban density, land-cover composition, sensor properties, and annotation characteristics. Previous work has explored domain adaptation, transfer learning, and geographically regularized learning strategies to improve robustness across domains [10], [11]. However, many of these approaches depend on access to target-domain data during training or adaptation, limiting their applicability in realistic deployment scenarios where annotated target data may not be available.

Besides structural domain differences, illumination variation and shadows increase the difficulty of segmentation in urban environments by altering spectral characteristics and obscuring object boundaries [12]. Shadowed regions can introduce ambiguity between vegetation, buildings, and artificial surfaces, especially in dense urban scenes. Although previous work has investigated shadow detection and compensation techniques, the relative contribution of shadows compared to broader domain shift factors remains not sufficiently understood.

Existing work primarily improves performance across cities through adaptation mechanisms requiring target-domain access. Source-only cross-city generalization remains comparatively less explored in aerial image segmentation due to variations in urban morphology, land-cover composition, and imaging characteristics [13]. In contrast, this study investigates whether failure patterns can be systematically characterized under source-only deployment conditions, with an emphasis on urban density, structural complexity, and illumination variation.

III. METHODOLOGY

This study investigates source-only cross-city generalization for semantic segmentation of aerial imagery. A model is trained on a labeled source domain and evaluated directly on unseen target domains without domain adaptation or fine-tuning. This setup evaluates the ability of the model to generalize across differences in urban structure, illumination

conditions, and data distribution. Model performance is evaluated using Intersection over Union (IoU) and pixel accuracy.

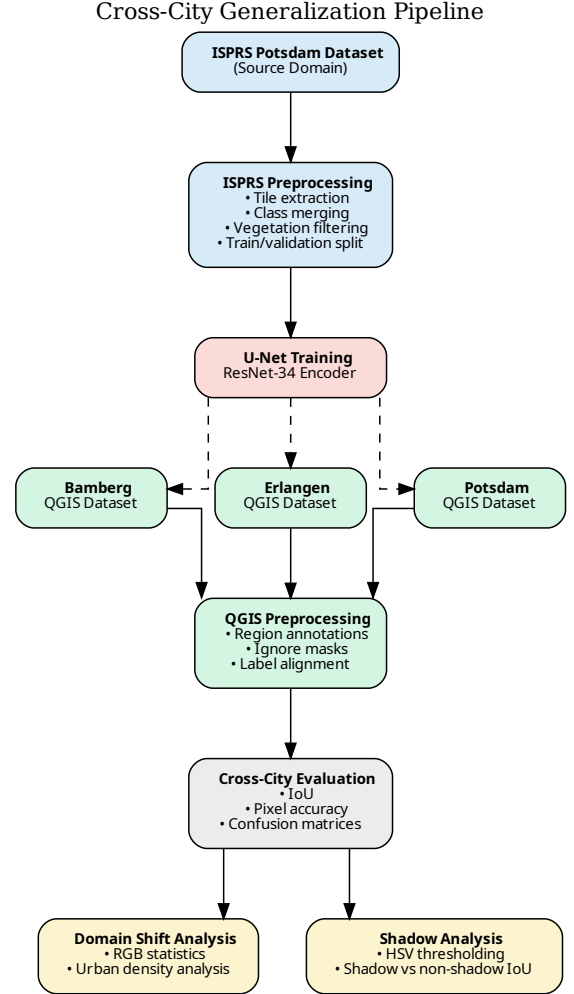


Fig. 1. Overview of the cross-city generalization pipeline. A U-Net model with a ResNet-34 encoder is trained on the ISPRS Potsdam source domain using vegetation-focused preprocessing and evaluated directly on unseen QGIS-derived target datasets from Bamberg, Erlangen, and Potsdam without domain adaptation. Separate preprocessing pipelines are applied to the source and target datasets before cross-city evaluation, domain shift analysis, and shadow analysis.

IV. DATASET AND PREPROCESSING

A. ISPRS Dataset

The ISPRS Potsdam dataset is used as the source domain for training and validation. It consists of high-resolution aerial RGB images with pixel-level annotations for urban land-cover classes.

For this study, the label space is simplified by merging related classes. The original ISPRS Potsdam dataset contains six semantic classes: background, impervious surfaces,

buildings, low vegetation, trees, and cars. For this study, the class configuration is modified to better align with the target application. The car class is merged with impervious surfaces to form a unified sealed surface category, as vehicles typically appear on artificial surfaces and are not directly relevant for vegetation analysis. After this modification, the dataset is reduced to five classes: background, sealed surface, building, low vegetation, and tree. During cross-domain evaluation, low vegetation and tree classes are further merged into a unified vegetation category to align with the annotation structure of the QGIS-derived datasets.

The original 6000×6000 ISPRS images are divided into 256×256 tiles using a sliding-window approach with a stride of 128 pixels, resulting in 50% overlap between adjacent tiles. This overlap increases sample diversity and improves spatial continuity near tile boundaries, while also ensuring sufficient representation of vegetation regions within the training data. The selected tile size balances spatial context and computational efficiency, as smaller tiles may not capture sufficient urban structure, whereas larger tiles increase memory requirements and scene complexity for U-Net-based segmentation. Tiles containing fewer than 50 tree pixels are excluded to reduce predominantly background-dominated samples while preserving sufficient vegetation representation for training. After preprocessing and filtering to retain vegetation-relevant regions, the dataset consists of approximately 41,000 training tiles and 3,800 validation tiles.

A group-wise 80–20 split based on the original parent images is used to prevent spatial data leakage between the training and validation sets. Despite preprocessing, the dataset remains dominated by sealed surface and vegetation regions, while building areas occupy comparatively fewer pixels.

B. QGIS-Derived Target Datasets

The target datasets were constructed from Google Satellite imagery acquired at different time periods and manually annotated using QGIS, which was also used for tile extraction and dataset preparation.

The target-domain datasets consist of aerial imagery from the cities of Bamberg, Erlangen, and Potsdam. Unlike the ISPRS Potsdam dataset, which contains winter imagery with dense pixel-wise semantic annotations, the QGIS-derived datasets primarily contain summer imagery captured under brighter illumination conditions and stronger vegetation visibility.

The QGIS annotations were originally created for vegetation monitoring and ecological analysis rather than dense semantic segmentation. Instead of fully labeling entire images, annotations were manually drawn around vegetation-relevant regions centered on individual trees and their surrounding urban context. Circular regions with approximately 15 m diameter were used to capture nearby vegetation, buildings, and sealed surfaces that can influence tree growth and environmental conditions.

Due to this annotation strategy, explicit separation between tree and low vegetation classes is not consistently available

in the QGIS-derived datasets. To maintain consistent semantic alignment across domains without introducing artificial labels, low vegetation and tree regions are merged into a unified vegetation category during evaluation. The final target-domain label space therefore consists of sealed surface, building, and vegetation classes.

All QGIS-derived tiles were extracted using the same input dimensions and approximately matching spatial scale as the ISPRS dataset to ensure consistent inference and metric computation across domains. Since QGIS annotations are already manually selected around vegetation-relevant regions, no extra filtering procedure is applied. Unlabeled regions outside annotated areas are assigned an ignore label and excluded from evaluation.

C. Label Alignment and Transformation

To enable consistent cross-domain evaluation, a unified label space consisting of sealed surface, building, and vegetation classes is defined in all datasets. The original ISPRS classes are mapped to this reduced label space by merging low vegetation and tree regions into a unified vegetation category, while cars are merged with sealed surfaces.

Since QGIS-derived datasets contain sparse region-based annotations rather than dense pixel-wise segmentation masks, pixels outside annotated regions are assigned an ignore label and excluded from evaluation. This ensures that performance metrics reflect predictions only within valid annotated regions and reduces evaluation bias caused by differences in annotation structure.

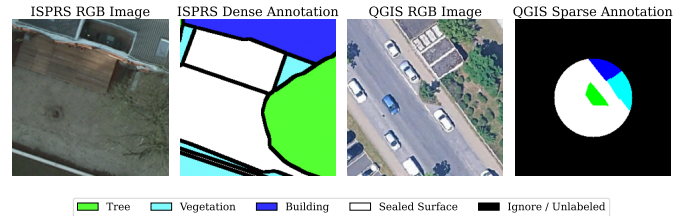


Fig. 2. The annotation differences between the ISPRS and QGIS-derived datasets. While the ISPRS dataset provides dense semantic annotations across entire images, the QGIS-derived datasets contain manually annotated vegetation-centered regions with unlabeled areas excluded during evaluation.

V. EXPERIMENTS AND EVALUATION

A. Model Architecture

A U-Net architecture with a ResNet-34 encoder pretrained on ImageNet is used for semantic segmentation. U-Net is selected due to its effective multi-scale spatial localization, computational efficiency, and stable performance on limited annotated datasets. Compared to more complex architectures such as transformer-based segmentation models, U-Net provides a simpler and computationally efficient baseline that is well suited for controlled analysis of cross-domain generalization behaviour.

A relatively lightweight ResNet-34 encoder is selected to balance representational capacity and computational complexity while maintaining robustness across varying urban scenes.

ImageNet pretraining provides strong low-level feature representations that improve generalization under cross-domain conditions.

The network is trained using the five-class ISPRS label configuration consisting of background, sealed surface, building, low vegetation, and tree classes.

B. Training Setup

The model was implemented using PyTorch and trained on 256×256 RGB image tiles using a batch size of 4. Training was performed using the AdamW optimizer with an initial learning rate of 1×10^{-4} . The combined Dice Loss and Cross-Entropy Loss were used to improve optimization stability and mitigate class imbalance during segmentation.

Data augmentation included random horizontal flipping, rotation, brightness adjustment, and contrast variation. These augmentations improved robustness against spatial and radiometric variability across urban environments and reduced overfitting to the source domain. ImageNet normalization was applied to all input images.

The model checkpoint that achieved the highest validation IoU was selected for evaluation. Training converged within 25 epochs without substantial further improvement on the validation set.

C. Evaluation Protocol

Model performance was evaluated using Intersection over Union (IoU) and pixel accuracy on both the ISPRS validation set and the QGIS-derived target datasets to assess cross-city generalization performance.

Cross-domain evaluation followed the merged vegetation evaluation setup described in Section IV-B to maintain consistency with the QGIS annotation structure. Pixels assigned to the ignore label in the QGIS-derived datasets were excluded from metric computation.

Per-class IoU scores were analysed to evaluate class-specific robustness across urban domains. Confusion matrices and row-normalized confusion matrices were used to identify dominant misclassification patterns between sealed surfaces, buildings, and vegetation regions.

D. Shadow Study Setup

To investigate the impact of shadows on segmentation performance, a shadow analysis was conducted on QGIS-derived datasets. Since explicit shadow annotations were not available, shadow regions were estimated using HSV-based thresholding. Pixels with normalized brightness (V) below 0.45 and saturation (S) below 0.35 were classified as shadow regions. Since manually annotated shadow masks were not available, the shadow regions represent heuristic approximations rather than precise ground-truth shadow annotations. The threshold values were empirically selected to approximate low-illumination regions while reducing inclusion of non-shadow dark surfaces.

Segmentation performance was evaluated separately for shadow and non-shadow regions using sealed-surface IoU and

pixel-wise error rate analysis. The error rate represents the proportion of incorrectly classified pixels within the evaluated region after excluding ignored pixels outside valid QGIS annotation masks. The analysis focuses on sealed surface segmentation, as shadowed regions commonly occur on roads and impervious surfaces where illumination changes frequently increase ambiguity with vegetation and building classes.

Moreover, the shadow analysis evaluates sealed-surface segmentation only within localized shadow and non-shadow regions rather than over the full image. Consequently, the resulting IoU values are intended for relative comparison between shadow conditions and are not directly comparable to the overall cross-city IoU values reported earlier. The evaluation therefore measures localized segmentation behaviour within shadow-conditioned regions rather than overall semantic segmentation performance across the full image.

The proportion of shadow pixels in each image was also computed to quantify shadow coverage across datasets. Although the shadow masks are approximate rather than manually annotated, they provide a consistent basis for comparative analysis across cities.

VI. RESULTS

A. ISPRS Baseline Performance

To establish baseline performance, the trained U-Net model was evaluated on the ISPRS validation set. This evaluation provides a reference for assessing cross-city generalization performance on the QGIS-derived datasets. Performance was measured using Intersection over Union (IoU) and pixel accuracy.

The model achieved strong segmentation performance on the ISPRS validation dataset, with a mean IoU of 0.719 and pixel accuracy of 0.848. Per-class IoU values are summarized in Table 1. The results indicate strong performance across

TABLE I
ISPRS BASELINE PERFORMANCE

Class	IoU
Background	0.4366
Sealed Surface	0.7993
Building	0.8850
Low Vegetation	0.7090
Tree	0.7650
Mean IoU	0.7190
Pixel Accuracy	0.8481

most classes, with high accuracy for building and sealed surface segmentation. Vegetation-related classes also achieved good performance, though slightly lower than artificial surface classes. Background segmentation showed comparatively lower performance, which is expected due to its heterogeneous composition and class imbalance.

These results demonstrate that the model successfully learned meaningful spatial features from the ISPRS dataset and achieved strong performance when evaluated on data drawn from the same distribution.

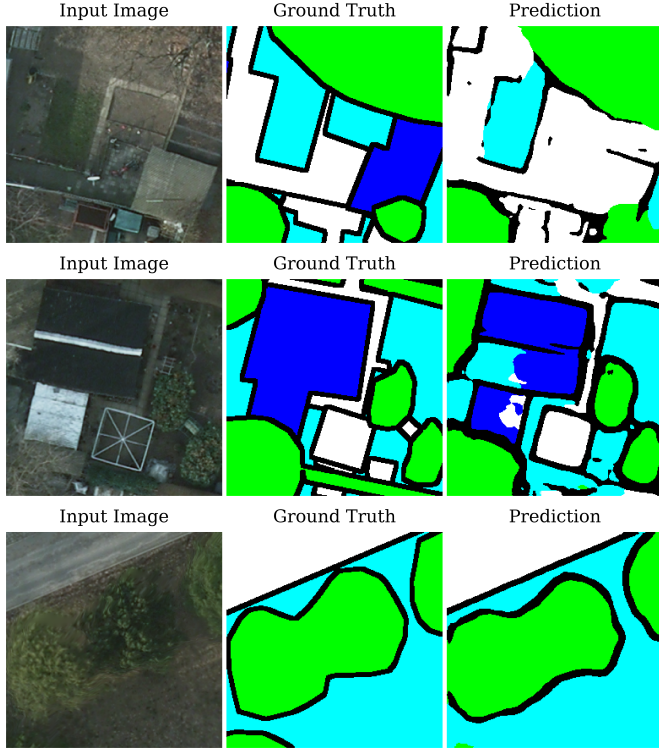


Fig. 3. Qualitative segmentation results on the ISPRS validation dataset. From left to right: input RGB image, ground truth mask, and model prediction.

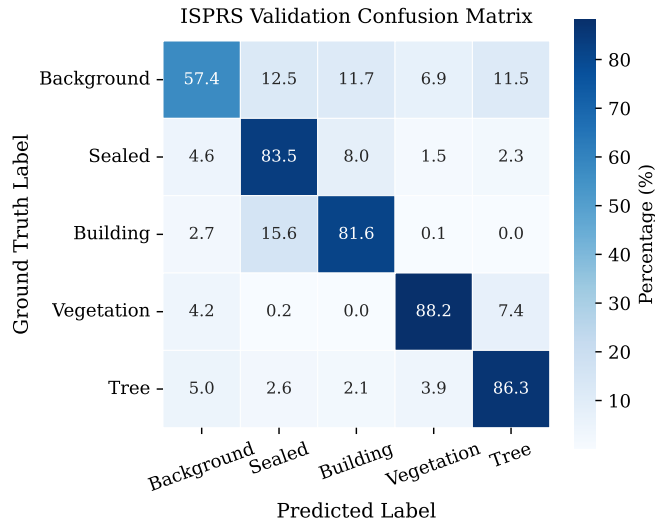


Fig. 4. Row-normalized confusion matrix for the ISPRS validation dataset. Strong diagonal responses indicate robust in-domain segmentation performance, while moderate confusion persists between buildings and sealed surfaces.

B. Cross-City Generalization Analysis

To evaluate domain generalization, the model trained on the ISPRS Potsdam dataset was tested on three unseen QGIS-derived cities: Bamberg, Erlangen, and Potsdam. These datasets differ from the training data in annotation style, urban layout, and illumination conditions. No fine-tuning or domain adaptation was performed, allowing direct assessment of cross-city generalization performance. Table II summarizes the performance across cities, followed by per-city analysis.

TABLE II
CROSS-CITY GENERALIZATION PERFORMANCE

City	Sealed	Building	Vegetation	Mean IoU	Pixel Accuracy
Bamberg	0.521	0.133	0.675	0.443	0.730
Erlangen	0.420	0.051	0.444	0.305	0.566
Potsdam	0.524	0.212	0.251	0.329	0.575

1) *Bamberg*: The model achieved the best cross-city performance on the Bamberg dataset, with a mean IoU of 0.443 and pixel accuracy of 0.730. Vegetation segmentation performed strongly with an IoU of 0.675, while sealed surface segmentation achieved moderate performance with an IoU of 0.521. Building segmentation showed considerably lower performance, with an IoU of 0.133.

The confusion matrix indicates strong confusion between buildings and sealed surfaces, with both missed building detections and overprediction of building regions. The relatively small number of building pixels in the ground truth further contributed to the lower building performance.

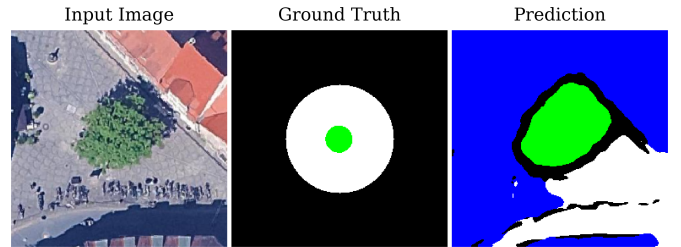


Fig. 5. Example of building misclassification in the Bamberg dataset. The model confuses buildings with surrounding sealed surfaces, resulting in inaccurate boundary delineation.

Despite this, vegetation segmentation remained stable, with high precision (0.820) and recall (0.792). This suggests that vegetation features learned from the ISPRS dataset generalized reasonably well to Bamberg. Overall, Bamberg exhibited the strongest observed cross-city performance among the evaluated target domains, although the limited dataset size (22 tiles) restricts the statistical weight of this comparison.

2) *Erlangen*: The model performance degraded substantially on the Erlangen dataset, with a mean IoU of 0.305 and pixel accuracy of 0.566. Building segmentation was particularly affected, achieving a very low IoU of 0.051. Sealed surface segmentation achieved an IoU of 0.420, while vegetation segmentation achieved an IoU of 0.444. Although vegetation

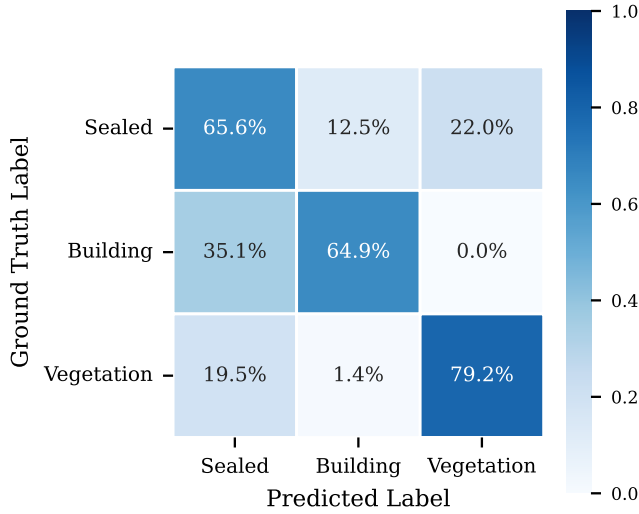


Fig. 6. Row-normalized confusion matrix for the Bamberg dataset.

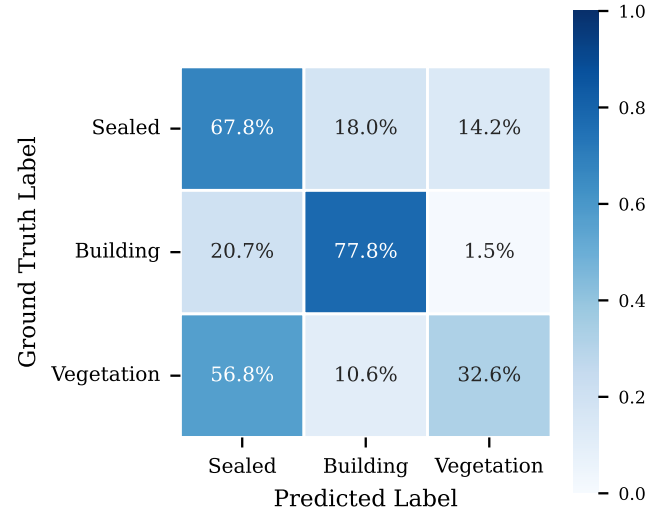


Fig. 8. Row-normalized confusion matrix for the QGIS-derived Potsdam dataset.

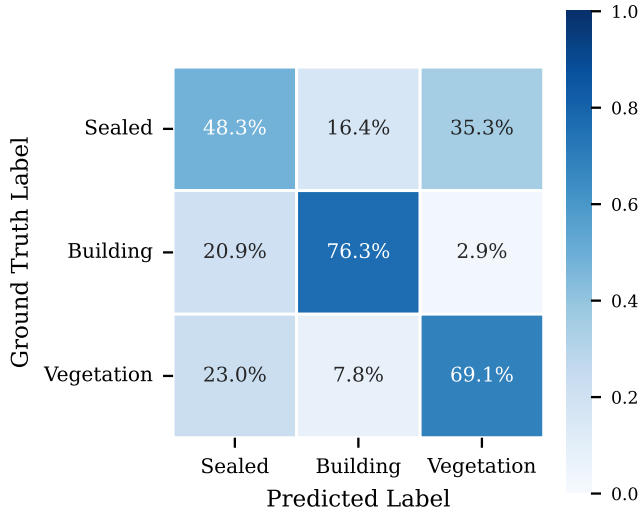


Fig. 7. Row-normalized confusion matrix for the Erlangen dataset.

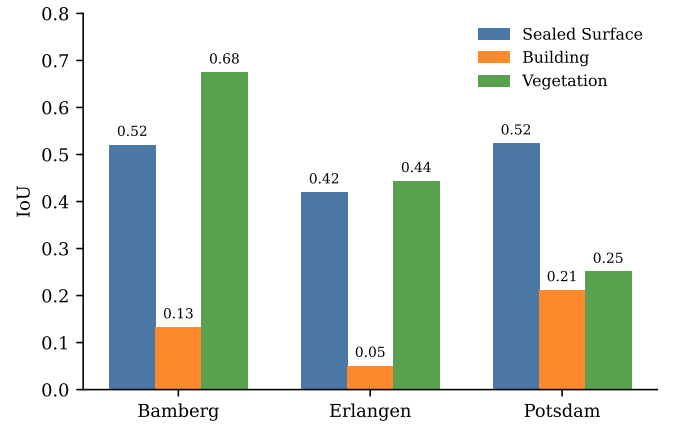


Fig. 9. Cross-city segmentation performance comparison across Bamberg, Erlangen, and Potsdam. Vegetation and building segmentation exhibit larger performance variation across cities than sealed surfaces.

performance remained moderate, confusion between vegetation and sealed surfaces increased compared to Bamberg.

The confusion matrix shows that buildings were frequently predicted but with very low precision (0.052) and high recall (0.763), indicating that the model over-predicted building regions.

3) *Potsdam*: Performance on the QGIS-derived Potsdam dataset showed moderate degradation compared to Bamberg but improved building segmentation compared to Erlangen. The model achieved a mean IoU of 0.329 and pixel accuracy of 0.575.

Sealed surface segmentation remained stable with an IoU of 0.524, while building segmentation improved to 0.212. However, vegetation segmentation performance dropped notably, with an IoU of 0.251.

The confusion matrix shows increased confusion between vegetation and sealed surfaces, specially in dense urban regions. Vegetation recall dropped to 0.326, indicating frequent confusion between vegetation and sealed surfaces.

C. Domain Shift Analysis

The cross-city evaluation results indicate a noticeable performance degradation when the model trained on ISPRS Potsdam was applied to unseen QGIS-derived datasets. To better understand this performance drop, domain shift between datasets was analysed using RGB statistics and class-wise prediction behaviour.

1) *RGB Distribution Differences*: To quantify radiometric differences between datasets, RGB mean and standard deviation values were computed for each city. Figure 10 summarizes the dataset statistics. The RGB statistics reveal no-

ticeable radiometric differences across the three cities. Erlangen exhibits higher green-channel intensity, suggesting denser vegetation or different illumination conditions. Potsdam shows larger standard deviation values, indicating greater variability in pixel intensities, which is consistent with more heterogeneous urban scenes. Bamberg displays comparatively lower variation, suggesting more uniform land-cover distribution.

These differences in colour distribution and illumination conditions contribute to domain shift, as the model was trained on ISPRS imagery with different radiometric characteristics.

2) *Class-wise Performance Degradation*: Domain shift effects are also visible in class-wise performance differences. Vegetation segmentation remained relatively stable in Bamberg but degraded remarkably in Potsdam and Erlangen.

- Bamberg showed strong vegetation performance (IoU: 0.675)
- Erlangen showed moderate vegetation performance (IoU: 0.444)
- Potsdam (QGIS) showed significant degradation (IoU: 0.251)

Building segmentation performance also varied strongly across cities. Erlangen exhibited the lowest building IoU (0.051), indicating strong domain shift effects in artificial surface representation.

3) *Prediction Distribution Analysis*: Prediction distributions highlight domain shift effects. In all three cities, sealed surface predictions dominated, suggesting that the model tended to default to artificial surface classes in uncertain regions. This behaviour was noticeable in Erlangen and Potsdam, where building and vegetation boundaries were frequently misclassified.

Building recall remained relatively high while precision was low, particularly in Erlangen. This indicates that the model predicted building regions aggressively but inaccurately.

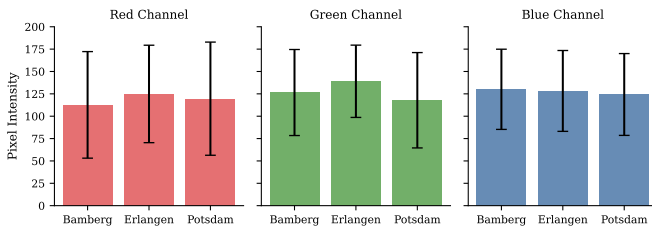


Fig. 10. Comparison of dataset-level RGB statistics across Bamberg, Erlangen, and Potsdam. Error bars represent standard deviation of pixel intensities across RGB channels.

D. Shadow Study Results

To investigate the impact of shadows on segmentation performance, a shadow analysis was conducted across all three QGIS-derived datasets: Bamberg, Erlangen, and Potsdam. Shadow regions were estimated heuristically.

Performance was evaluated separately within shadow and non-shadow regions using sealed-surface IoU and pixel-wise

TABLE III
SHADOW VS NON-SHADOW PERFORMANCE ACROSS CITIES

City	Shad. IoU	Non-Shad. IoU	Shad. Err.	Non-Shad. Err.	Shad. %
Bamberg	0.101	0.275	0.253	0.296	6.15
Erlangen	0.127	0.302	0.518	0.451	5.76
Potsdam	0.261	0.439	0.511	0.414	14.28

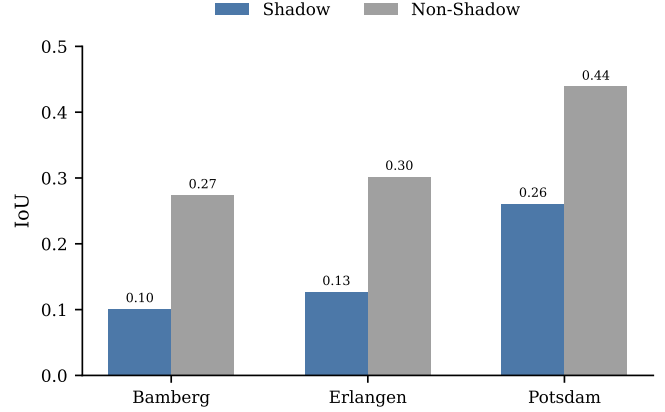


Fig. 11. Comparison of sealed-surface IoU within shadow and non-shadow regions across cities. Non-shadow regions consistently achieve higher IoU values, indicating that shadows introduce measurable localized degradation in sealed-surface segmentation performance.

error rate analysis. Evaluation was restricted to valid annotated regions only, excluding ignored pixels outside the QGIS annotation masks.

The analysis focused primarily on sealed-surface segmentation, since shadows frequently occur on roads, pavements, and artificial surfaces where illumination changes can increase ambiguity with vegetation and surrounding urban structures. Table III summarizes the shadow analysis results across all cities.

1) *Bamberg*: For Bamberg, shadow regions accounted for approximately 6.15% of total pixels. Sealed-surface IoU within shadow regions was 0.101, compared to 0.275 in non-shadow regions. This indicates noticeable localized degradation in segmentation performance under shadow conditions.

2) *Erlangen*: Erlangen exhibited similar behaviour, with shadow coverage of approximately 5.76%. Sealed-surface IoU within shadow regions was 0.127, while non-shadow regions achieved a higher IoU of 0.302. Shadow-region error was also slightly higher than non-shadow error, indicating increased difficulty in shadowed urban regions.

3) *Potsdam*: Potsdam showed the highest shadow coverage at approximately 14.28%. Sealed-surface IoU within shadow regions reached 0.261, while non-shadow regions achieved 0.439. Although shadow regions consistently reduced segmentation quality, performance degradation was also considerable in non-shadow regions due to the overall structural complexity of the dataset.

Across all cities, non-shadow regions consistently achieved

higher IoU values than shadow regions, indicating that shadows contribute to localized segmentation degradation. However, substantial errors also remained outside shadowed areas, suggesting that broader domain-shift factors such as urban density, structural complexity, annotation sparsity, and seasonal variation remain major contributors to cross-city generalization failure.

VII. DISCUSSION

A. Cross-City Performance Interpretation

The cross-city evaluation results demonstrate a clear degradation in segmentation performance when the model trained on ISPRS Potsdam is applied to unseen urban environments. However, this degradation is not uniform across cities and can be explained through measurable factors such as urban density, class distribution, and radiometric differences.

Bamberg exhibits the strongest observed cross-city performance, particularly for vegetation segmentation. However, the Bamberg dataset contains only 22 annotated tiles, which limits the statistical strength of this comparison. Despite this limitation, the results still suggest that lower urban density contributes to improved vegetation segmentation performance. As shown in Figure 12, tiles with low sealed surface ratio (i.e., less urbanized regions) consistently achieve a higher vegetation IoU, often exceeding 0.8. This trend indicates that the model generalizes more effectively in simpler environments where vegetation is spatially distinct and less fragmented.

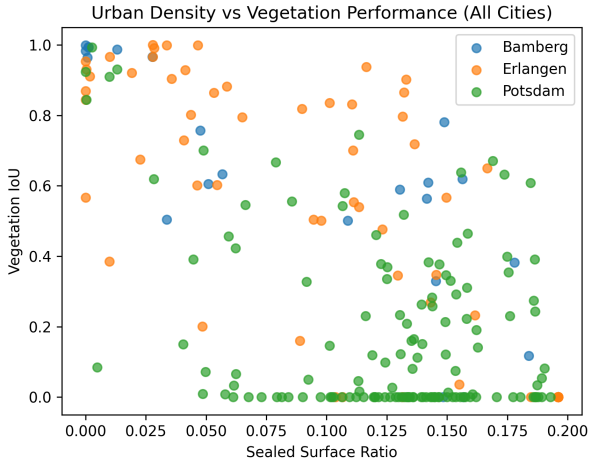


Fig. 12. Relationship between sealed surface ratio and vegetation IoU across Bamberg, Erlangen, and Potsdam.

In contrast, Erlangen shows a significant drop in performance, especially for building segmentation. The scatter plot in Figure 12 shows that as sealed surface ratio increases, vegetation IoU becomes highly unstable, with many samples collapsing toward zero. This suggests that in moderately urbanized regions, the model struggles to distinguish between vegetation and artificial surfaces. Additionally, the boxplot in Figure 13 confirms that urban tiles have substantially lower vegetation IoU compared to rural tiles, reinforcing

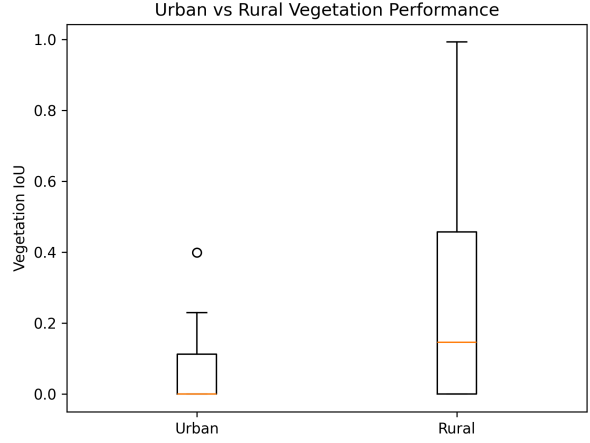


Fig. 13. Comparison of vegetation segmentation performance in urban and rural regions. Rural areas consistently achieve higher vegetation IoU, indicating that dense urban environments increase segmentation difficulty.

the impact of urban density on segmentation performance. Potsdam presents the most challenging case despite being geographically similar to the source dataset. Although building segmentation improves compared to Erlangen, vegetation performance drops significantly. This trend is evident in Figure 14, where Potsdam shows a lower median vegetation IoU and a wider distribution of low-performing samples. Also, Figure 12 reveals that many high-density tiles (higher sealed surface ratios) correspond to near-zero vegetation IoU, indicating severe confusion in dense urban environments.

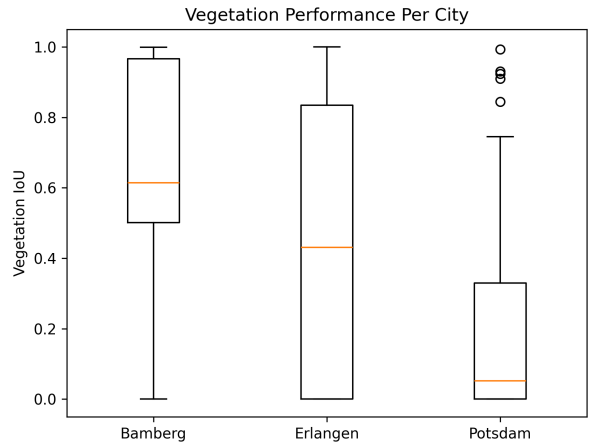


Fig. 14. Distribution of vegetation IoU across Bamberg, Erlangen, and Potsdam. Potsdam exhibits a larger proportion of low-performing vegetation samples compared to Bamberg and Erlangen.

Across all cities, a consistent pattern emerges: increasing urban density leads to degraded vegetation segmentation performance. This relationship is quantitatively supported by the negative correlation between sealed surface ratio and vegeta-

tion IoU in Figure 12, as well as the clear separation between urban and rural performance distributions in Figure 13. These results confirm that the model is sensitive to spatial complexity and struggles in environments with mixed land-cover types.

Besides urban density, class distribution plays a significant role. The ground truth statistics show that building pixels are relatively scarce compared to sealed surfaces and vegetation, notably in Erlangen. This imbalance contributes to unstable building predictions and increased misclassification. Furthermore, RGB statistics indicate noticeable radiometric differences across cities, suggesting that variations in illumination, colour distribution, and seasonal appearance contribute to domain shift as well. In particular, the ISPRS source imagery primarily contains winter scenes, whereas the QGIS-derived target datasets consist largely of summer imagery with denser vegetation and brighter illumination conditions. These seasonal differences likely introduce additional spectral and textural variation that affects cross-city generalization performance.

Overall, the results indicate that domain shift in aerial image segmentation is driven by a combination of structural complexity, class imbalance, and radiometric variation. Among these, urban density emerges as the dominant factor influencing generalization performance, with denser and more heterogeneous environments leading to significantly higher segmentation errors.

B. Failure Modes

The cross-city evaluation reveals several consistent and systematic failure modes that highlight the limitations of the model when applied to unseen urban environments. These errors are primarily driven by class ambiguity, dataset imbalance, and increasing scene complexity.

The most prominent failure mode is the confusion between buildings and sealed surfaces. As observed in the confusion matrix (Figure 7, Figure 8), building predictions achieve high recall, while precision remains low, indicating systematic overprediction and confusion with sealed surfaces. This leads to low building IoU despite relatively high recall, notably in Erlangen. The model tends to associate artificial textures such as rooftops, roads, and pavements with a common representation, making it difficult to distinguish between structurally similar classes.

A second major failure mode is the misclassification of vegetation in urban regions. While vegetation segmentation performs well in low-density areas, performance degrades significantly in dense environments. As previously shown in Figure 12, many high-density tiles correspond to near-zero vegetation IoU. This indicates that vegetation is frequently misclassified as sealed surface when surrounded by artificial structures, resulting in reduced recall and fragmented predictions.

The model also exhibits a clear class bias toward sealed surfaces, which dominate predictions in uncertain regions. This behaviour is supported by the prediction distributions, where sealed surfaces consistently account for the largest number of

predicted pixels across cities. This bias is likely influenced by the prevalence of large homogeneous regions in the training data, causing the model to default to sealed surface predictions when uncertainty is high.

Another observed limitation is the model’s difficulty in capturing small and irregular structures, specially buildings. Smaller buildings or those with complex shapes are often missed or only partially segmented. This suggests that the model relies more on coarse spatial patterns rather than fine-grained structural features, limiting its ability to generalize to diverse urban layouts.

C. Role of Shadows

The shadow analysis was conducted to evaluate whether illumination differences contributed to the performance degradation observed in cross-city generalization.

Across all three cities, sealed-surface IoU within shadow regions was consistently lower than within non-shadow regions, indicating that shadows introduce measurable localized segmentation difficulty (Figure 11).

Although shadows negatively affected segmentation quality, substantial errors were also observed outside shadow regions. In many cases, segmentation failures occurred in structurally complex urban scenes containing fragmented vegetation, roads, buildings, and heterogeneous artificial surfaces. This suggests that shadows are a secondary contributing factor, but not the sole driver of cross-city performance degradation.

The heuristic shadow masks revealed that many shadowed regions overlap with dense vegetation and complex urban boundaries, increasing ambiguity between vegetation, sealed surfaces, and buildings. Also, the sparse circular annotation structure of the QGIS-derived datasets makes localized IoU measurements highly sensitive to small boundary disagreements.

In general, the results suggest that shadows contribute to localized degradation in sealed-surface segmentation performance. However, broader domain-shift factors, such as urban density, structural complexity, annotation sparsity, and seasonal radiometric variation, remain dominant challenges for cross-city generalization.

VIII. CONCLUSION

This study investigated source-only cross-city generalization for urban vegetation segmentation using aerial imagery. A U-Net model with a ResNet-34 encoder was trained on the ISPRS Potsdam dataset and evaluated directly on unseen QGIS-derived datasets from Bamberg, Erlangen, and Potsdam without domain adaptation or fine-tuning. Although the model achieved strong in-domain performance, segmentation accuracy degraded massively across unseen urban environments, highlighting the challenges of cross-city domain transfer.

The results demonstrate that urban density and structural complexity are the main factors that influence generalization performance. Vegetation segmentation degraded in dense urban regions with heterogeneous land-cover composition, while sealed surface segmentation remained comparatively

stable across cities. Building segmentation showed inconsistent behaviour due to the class imbalance and a strong confusion with the surrounding artificial surfaces. Failure analysis also revealed systematic prediction biases towards sealed surfaces and reduced robustness in complex urban layouts.

Shadow analysis showed that illumination variation contributes to localized degradation in sealed-surface segmentation performance, with shadow regions consistently achieving lower IoU values than non-shadow regions. However, substantial segmentation errors also remained outside shadowed areas, indicating that broader domain-shift factors such as urban density, structural complexity, annotation sparsity, and seasonal variation remain the main contributors to cross-city generalization failure.

A limitation of this study is the relatively small size and sparse annotation structure of the QGIS-derived target datasets, as well as the seasonal mismatch between the ISPRS source imagery and the QGIS-derived target imagery. Since the ISPRS dataset mainly contains winter scenes while the target datasets largely contain summer imagery, part of the observed performance degradation may be influenced by seasonal domain shift alongside structural and urban-density differences. Future work should investigate domain generalization strategies that improve robustness to dense urban environments and seasonal variation, including multi-city training, stronger class balancing approaches, and architectures capable of better separating vegetation from structurally similar artificial surfaces.

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